EVAN T.C. QUINTANA

+1 (760) 634-4445 | etc.qqq@gmail.com | github.com/evanforreal | linkedin.com/in/etcqqq

SUMMARY

My professional directions lie in formal verification methods for {secure inference, loss-of-control response scenarios, model integrity, multilateral agreements like international compute pacing} and my motivations draw me towards those angles in AI sec which are severely underworked but critical. Right now my burning interest is in how developments on the frontier affect and/or rush decisions by downstream actors like open-source or -weight labs.

EXPERIENCE

R&D Data Scientist, AI Security | Sandia National Laboratories, Org. 5563 (Robust Systems Analysis) | *10/26 – current*

- Focuses: counter-adversarial ML, formal methods for verification, generative AI for national security. Q+SCI clearance in process.

Graduate R&D Intern | Sandia National Laboratories, Org. 5565 (Math & Analytics for natsec) | *06/26 – 09/26*

- Post-trained an 8B model in a simulated adversarial setting to characterize the possibilities for compromise of safety-tuned base model snapshots. Held role of co-lead on the AI red team within the evaluations group of a division-1400 trilab genAI project.
- Research directions: AI integrity, adversarial robustness, verification of secure compute.

Teaching Assistant, Department of Applied Mathematics | University of California, Santa Cruz | *01/26 – 06/26*

Intern, Global Engagement Center, China Desk (GEC) | U.S. Department of State | *2024*

- Authored 10+ internal reports on malign disinformation campaigns (example country 1: analyzed 600+ national outlets for content-sharing from confirmed global propaganda newswires), distributed to U.S. embassies. Held Secret clearance.

TECHNICAL PROJECTS

Scientific Machine Learning (SciML) | graduate course projects

- Conditional diffusion for sparse-to-full field reconstruction
- Conditional variational autoencoder (CVAE) for autoregressive probabilistic forecasting

Echo-state networks for data-driven superparameterization | undergraduate research

Election disinformation in Taiwan via network analysis | Fulbright semifinalist, open research grant

- Using a 700k-post corpus of social media interactions, I performed community clustering and network analysis to isolate an activity spike around (for example) a false narrative that presidential candidate Lai Ching-Te had an illegitimate child and was unfit for office. RAND research independently identified the same narrative two years later.

AI Red-Teaming Exercise: Catastrophic Risk | Expert participant, Nemesys Insights (under NDA)

EDUCATION

University of California, Santa Cruz: M.S. Scientific Computing and Applied Mathematics, 06/26. GPA 4.0.

University of California, Santa Cruz: B.S. Applied Mathematics, summa cum laude, and B.A. Global Economics. GPA 3.77.

National Taiwan University: Study abroad. Focus: Chinese, political economy of semiconductors.

Graduate coursework: SciML, numerical linear algebra, numerical optimization, applied optimal control, stochastic differential equations, high-performance computing; independent study in uncertainty quantification for SciML.

SKILLS AND CLEARANCE

Technical: Python (pytorch, jax, scipy, xarray, vllm), Rust, Git, Linux.

Languages: English (native), Chinese (proficient), Spanish (proficient).

Clearance: Q+SCI, in progress as of October 26. **Interim Secret**, active from Jan 24 to March 24.

Things that worry me: How do we engineer reliable SIDCs? What paradigm will succeed RL to prevent inner misalignment? What multilateral offramps do we have towards pacing the frontier?

TALKS

UCSC Digital Scholarship Symposium: Gave three-minute lightning talk. Presented motivations, methods, and findings for Taiwan disinformation project.

Koret Research Symposium: Gave a year-end poster session regarding the abovementioned Taiwan disinformation project. Presented findings around core activity spikes identified and possible future work in the field.